piBrick PocketCM5

Thank you for choosing piBrick!

Welcome to your piBrick PocketCM5. Use this guide to assemble your device, complete the first boot, install the Linux drivers, and find useful project resources.

⚙ Quick Driver Installation Guide

Open a terminal on your piBrick, or connect to it over SSH, and run these commands:

```
cd ~
git clone https://github.com/amarullz/pibrick-drivers.git
cd pibrick-drivers
bash install.sh
```

Follow the installer's prompts, and restart your piBrick if requested.

🧩 Assembly Guide

1. Open the enclosure by following the *Case Opening Guide*.
2. Fit the Raspberry Pi Compute Module 5 (CM5) onto the piBrick Mainboard.
3. Use the supplied Hextool to install the hex spacer from the heatsink.
4. Position the thermal pads, then secure the heatsink with the screws supplied in its box.
5. Unscrew and remove the battery holder.
6. Position the battery and reinstall the holder.
7. Connect the battery cable to the connector at the bottom-left of the piBrick Mainboard.
8. Close the enclosure, following the *Case Opening Guide* if needed.

💾 First Boot and Driver Installation

1. Flash your chosen operating system image to a microSD card.
2. Insert the microSD card into your piBrick.
3. Power on your piBrick.
4. Install the Linux drivers using SSH, an HDMI-connected display and keyboard, a serial console, or Raspberry Pi Connect. See the *Quick Driver Installation Guide* and the piBrick GitHub page for details.

🔗 piBrick Links and Resources

- 🌏 **Website:** amarullz.com
- 🔧 **piBrick GitHub:** github.com/amarullz/pibrick
- 🛟 **piBrick Discord:** discord.gg/e9qUdJ4aVr
- 🛍️ **Tindie Store:** tindie.com/stores/amarullz
- 🔬 **OSHWLab (PCB and Schematic):** oshwlab.com/amarullz/pibrick-pocketcm5
- 📺 **YouTube:** youtube.com/@amarullz
- ⚓ **Linux Driver:** github.com/amarullz/pibrick-drivers
- ⌨ **Keyboard Firmware:** github.com/amarullz/pibrick_pocketcm5_keyboard

🤖 **piBrick GitHub**
Scan for project source

🛟 **piBrick Discord**
Scan to join the community

🔪 Opening the Case

Scan the QR code or open the link to watch the case-opening video before disassembling your piBrick.

Case Opening Guide
youtube.com/watch?v=LbCXG7IgLO4
Scan to watch the video.

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Package Guide & Safety

Package contents, assembly hardware and product information

📦 Package Contents

Check the included parts before beginning assembly.

- **Main Unit**
- **3D-printed parts:**
  hex tool, fan enclosure, antenna holder and switch spare part

- **CM5 Heatsink**
  (PWM Fan, thermal pads, screws and spacers included)
- JST2-PH and JST4-SH with cable

- Dual-band Wifi Antenna
- Screw Sets - see **Screw Reference**
- Quick user manual

Not included: Raspberry Pi Compute Module 5, LiPo battery and M.2 NVMe SSD.

🔩 Screw Reference

Size	Purpose
M2.0 × 10 mm	Main casing fasteners
M1.7 × 5 mm	Battery holder, Speaker-IO and display/keyboard holder
M2.0 × 3 mm	M.2 NVMe SSD fastener
M1.7 × 8 mm	Fan enclosure

Use the correct screw size for each mounting point. Do not overtighten screws or force them into place.

⚠️ Important Safety Information

⚠️ WARNING: Read all instructions before assembling or operating this product. Failure to follow these instructions may result in equipment damage, fire, electric shock, or personal injury.

⚡ Assembly & ESD Precautions

Install or remove the CM5 only with power disconnected. Use appropriate ESD precautions. Fully seat connectors and flex cables without forcing or sharply bending them.

🔋 Battery Safety

Battery not included. Use only a high-quality 1S LiPo battery with integrated PCM/BMS protection. Never use a damaged or swollen battery, reverse its polarity, or short its terminals. Disconnect immediately if excessive heat, smoke or unusual odor occurs. Follow the battery manufacturer's instructions.

🚨 Intended for technically competent users. Assemble and operate this kit in accordance with this manual.

✅ Product Identification

Product Name	piBrick Pocket-CM5
Hardware Revision	Version 1.1 REV-5
Product Type	Portable carrier board kit for Raspberry Pi CM5
Compatible Module	Raspberry Pi Compute Module 5 (sold separately)
Power Source	Single-cell (1S) LiPo battery (sold separately)
Nominal Input Voltage	3.7 V DC
Maximum Internal Voltage	7.0 V DC (display boost rail)
Intended Use	Open-source portable carrier board kit designed for use with the Raspberry Pi Compute Module 5.

🕹️ Key Features

🖥️ Display
3.92-inch AMOLED touchscreen · 1080 × 1240 · 90 Hz

Computing & Storage
Raspberry Pi CM5 · microSD/eMMC · M.2 NVMe SSD support

🖱️ Controls
Physical QWERTY keyboard with integrated trackpad, rotary control and user buttons

🛜 Connectivity
USB-A/USB-C ports, full-size HDMI, micro-HDMI and 3.5 mm audio

🔈 Audio & Sensors
Stereo speakers, audio amplifier, microphone and accelerometer

💊 Expansion
Internal USB, I²C and GPIO expansion connectors

CM5, battery and NVMe SSD are separate components and are not included with this kit.

⌨️ Keyboard & Trackpad Quick Tips

Symbols & numbers
Press **SYM** for the symbol layer. Tap for one-shot input; hold to stay on the layer.

Extended keys
Double-tap **SYM** for Layer 2. Tap **SYM**, then **Right Shift**, for Layer 3.

Trackpad / arrow mode
Tap and hold **BACK** to switch modes. A blinking indicator means arrow mode; steady means mouse mode.

Keyboard customization
Use **https://vial.rocks/** to customize key mappings. Firmware and recovery keymap are available in the keyboard firmware repository.



WEEE Disposal
Do not dispose of this product with household waste. At the end of its service life, take it to an appropriate WEEE collection or recycling facility in accordance with local regulations and Directive 2012/19/EU.

Community Support: **https://discord.gg/e9qUdJ4aVr** | Contact: **amarullz@gmail.com**